# ONLINE FOOD DELIVERY MOBILE APP

A smart, convenient and real-time platform connecting customers, restaurants and delivery partners

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# 1. Introduction

- Online food delivery apps allow customers to discover restaurants, order food and pay digitally from one platform.
- They connect three key stakeholders: customers, restaurants and delivery partners.
- Real-time order tracking, digital payments and personalized recommendations improve convenience.
- The proposed app focuses on a simple user experience, reliable order management and efficient delivery coordination.

# 2. Problem Statement

- Customers may face long waiting times, unclear order status and limited delivery visibility.
- Restaurants need an efficient way to manage menus, incoming orders and availability.
- Delivery partners require accurate order details, optimized routes and status updates.
- Payment failures, incorrect orders and poor communication can reduce customer satisfaction.
- Therefore, a unified mobile platform is needed to manage the complete food-ordering lifecycle.

# 3. Objectives

- Provide a simple and user-friendly mobile interface for browsing and ordering food.
- Enable secure registration, login and digital payment.
- Allow restaurants to manage menus, prices, availability and incoming orders.
- Provide delivery partners with order assignment, navigation and delivery-status updates.
- Offer real-time notifications and order tracking to improve transparency.
- Maintain reliable data, secure transactions and scalable backend services.

# 4. Literature Review

| Area | Common Approach | Key Benefit | Opportunity for Proposed App |
|---|---|---|---|
| Mobile ordering | Restaurant discovery + cart | Fast ordering | Simpler flow and better personalization |
| Digital payment | Cards, wallets, UPI | Cashless checkout | Reliable payment confirmation |
| GPS tracking | Live delivery location | Visibility | Accurate ETA and status updates |
| Recommendation | History/popular items | Personalization | Context-aware food suggestions |

# 5. System Modules

### 1 User & Authentication Module

Registration, login, profile, address management and role-based access.

### 2 Restaurant & Menu Module

Restaurant listing, menu browsing, search, filters, item availability and offers.

### 3 Order & Payment Module

Cart, order placement, payment, order confirmation, invoices and notifications.

### 4 Delivery & Tracking Module

Delivery assignment, GPS location, route support, live status and proof of delivery.

# Module 1 — User & Authentication

- User registration using mobile number/email and secure password authentication.
- Profile management: name, contact details and saved delivery addresses.
- Role-based access for customer, restaurant administrator and delivery partner.
- Session/token management protects authenticated API requests.
- Validation prevents invalid or incomplete account information.

# Module 2 — Restaurant & Menu

- Displays nearby or searchable restaurants with cuisine, rating and delivery information.
- Restaurants can add, edit, remove and temporarily disable menu items.
- Customers can browse categories, search dishes and apply filters.
- Menu data includes item name, description, price, image and availability.
- Offers and recommendations can improve discovery and customer engagement.

# Module 3 — Order & Payment

- Customers add items to cart, review quantity and confirm delivery address.
- Order service calculates item totals, taxes, delivery charges and discounts.
- Payment gateway handles digital payment and returns transaction status.
- Order states can progress through: Placed → Accepted → Preparing → Picked Up → Delivered.
- Push notifications keep customers and restaurants informed at each important stage.

# Module 4 — Delivery & Tracking

- Delivery partner receives available orders and accepts an assignment.
- GPS services support navigation from restaurant to customer location.
- Customers can view delivery status and estimated arrival time.
- Delivery partner updates status such as picked up, on the way and delivered.
- Completion can include OTP/proof of delivery and order history.

# 6. System Architecture

High-

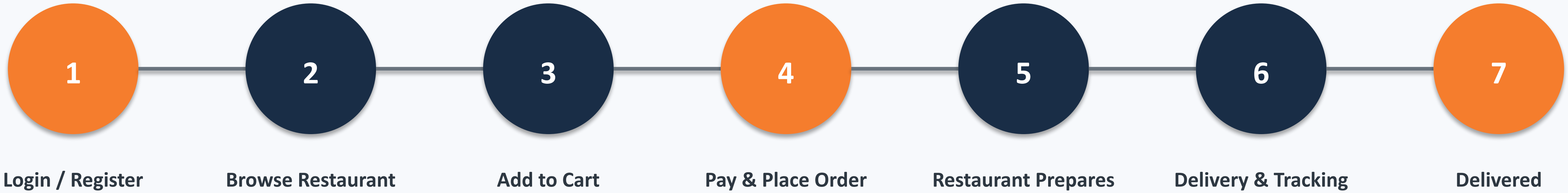
**Mobile App** Customer App | Restaurant App | Delivery Partner App

**Backend / API Layer** Authentication | Restaurant Service | Order Service | Delivery Service

**Data & Integration** Database | Payment Gateway | Maps/GPS | Push Notifications

**External Services** Cloud Hosting | SMS/Email | Analytics & Monitoring

# 7. Order Workflow



# 8. Future Scope

- AI-based personalized recommendations based on preferences and ordering history.
- Predictive ETA and intelligent delivery-route optimization using real-time traffic.
- Voice-based ordering and multilingual interfaces for wider accessibility.
- Cloud autoscaling and microservices for handling high traffic during peak hours.
- Advanced restaurant analytics for demand forecasting and inventory planning.
- Loyalty programs, subscriptions and smart promotional campaigns.
- Integration with smart kitchens, drones/robots and emerging last-mile delivery technologies.

# 9. Conclusion

- The proposed Online Food Delivery Mobile App provides an end-to-end digital ordering and delivery platform.
- The four modules work together to manage users, restaurants, orders/payments and delivery tracking.
- A secure backend, real-time updates and integrated services improve reliability and convenience.
- The architecture is scalable and can support future AI, cloud and intelligent delivery features.
- Overall, the system aims to reduce ordering effort, improve operational efficiency and enhance customer satisfaction.

# 10. References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# THANK YOU

Questions & Discussion